FARMGATE AI — GRAIN ASSAY CERTIFICATE

Low-Cost Smartphone Vision Assaying System | AGMARK Standards Compliance Assessment

Sample ID: SMP-2026-0841	**Date/Time:** 2026-10-19 17:59:33
Grain Type: Basmati / Extra Long	**Inspector:** Farmgate Field Assayer

ASSESSED QUALITY GRADE

REJECTION / SUB-STANDARD

Exceeds tolerance limits: Discoloured grains (54.5%) exceed 6.0% limit.

1. Morphometric & Optical Defect Breakdown

Quality Parameter	Measured Value	Standard AGMARK Limit	Status
Total Grains Assayed	44 kernels	Min 30 kernels required	Adequate
Whole Grains (Head Rice)	12 (27.3%)	>= 75.0% for Grade A/B	Warning
Broken Grains	5 (11.4%)	<= 5.0% (G-A) / <= 12.0% (G-B)	Pass
Discoloured / Damaged Grains	24 (54.5%)	<= 1.0% (G-A) / <= 3.0% (G-B)	Fail
Chalky Grains (Opacity >= 25%)	3 (6.8%)	<= 3.0% (G-A) / <= 6.0% (G-B)	Fail
Average Kernel Length	7.26 mm	Nominal: 7.3 mm	Recorded
Average Kernel Width	2.62 mm	Nominal: 2.6 mm	Recorded
Length-to-Breadth Aspect Ratio	2.95	>= 3.0 (Long Grain Standard)	Medium

2. Assaying Standard Compliance Methodology

Measurements conducted using Farmgate AI high-precision computer vision pipeline. Fiducial perspective rectification performed via OpenCV ArUco (DICT_4X4_50) Homography at sub-millimeter scale. Grain segmentation utilized watershed separation with Otsu-adaptive binarization. Grain dimensional and defect grading adheres to the Directorate of Marketing and Inspection (DMI), Ministry of Agriculture & Farmers Welfare, Government of India (AGMARK Milled Rice Grading and Marking Rules).

Automated Assaying Signature: FARMGATE-AI-VERIFIED-V1.0	**Quality Assurance Officer:** _____ (Sign/Seal)